

Uniform Homeomorphism of Separable Banach Spaces: Known E_{ℓ_∞} Lower Bound

Literature-implied partial status note

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Status. This is a *literature-implied answer (partial subcase)* for the open problem in Ferenczi–Louveau–Rosendal, arXiv:math/0610289. It is not a full solution of the analytic-completeness question.

Source question. In the section “Uniform homeomorphism of complete separable metric spaces,” after proving that uniform homeomorphism of complete separable metric spaces is complete analytic, Ferenczi–Louveau–Rosendal write that they have not been able to replace their metric spaces $P(X)$ by Banach spaces and ask:

What is the complexity, with respect to $\leq_B$, of uniform homeomorphism between separable Banach spaces?

In particular, they ask whether this relation is analytic complete.

Supporting theorem. Gao–Jackson–Sari, arXiv:0901.4092, explicitly identify this as the main problem left open from Ferenczi–Louveau–Rosendal. Their Theorem 1.1 constructs a Borel family $\{S_{\vec{x}} : \vec{x} \in \mathbb{R}^\omega\}$ of separable Banach spaces such that, for all $\vec{x}, \vec{y}$,

$$\vec{x} - \vec{y} \in \ell_\infty \iff S_{\vec{x}} \text{ and } S_{\vec{y}} \text{ are uniformly homeomorphic} \iff S_{\vec{x}} \text{ and } S_{\vec{y}} \text{ are locally equivalent.}$$

Equivalently,

$$E_{\ell_\infty} \leq_B E_{\text{UH}}^{\text{Banach}},$$

where $E_{\text{UH}}^{\text{Banach}}$ denotes uniform homeomorphism between separable Banach spaces. Their Theorem 1.2 further proves that local equivalence between separable Banach spaces is Borel bireducible with E_{ℓ_∞} .

Interpretation. This gives a nontrivial partial answer to the source problem: uniform homeomorphism of separable Banach spaces is at least complete K_σ in the Borel-reducibility hierarchy. The same paper explains that this lower bound is the strongest result obtainable from local finite-dimensional structure alone, via Ribe’s theorem and local equivalence.

Remaining open scope. The analytic-completeness part of the Ferenczi–Louveau–Rosendal problem is not answered by Gao–Jackson–Sari. They state that proving $E_0^\omega \leq_B E_{\text{UH}}^{\text{Banach}}$ would already show that uniform homeomorphism is strictly more complex than E_{ℓ_∞} and would move beyond the local-structure barrier.

References

- [1] V. Ferenczi, A. Louveau, and C. Rosendal, *The complexity of classifying separable Banach spaces up to isomorphism*, arXiv:math/0610289.
- [2] S. Gao, S. Jackson, and B. Sari, *On the complexity of the uniform homeomorphism relation between separable Banach spaces*, arXiv:0901.4092.